

EduGenie-Google Gemini Powered Learning Assistant

Government Arts and Science College-Manur-MSU142

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Project Design Phase

The **Project Design Phase** takes the broad business requirements gathered during initiation and transforms them into actionable blueprints, system architectures, and user interfaces before actual construction or coding starts.

1.High-Level Design (HLD):Macro-level architectural decisions.

Establish the overall framework and structural roadmap for the solution.

- **System Architecture:** Select the technology stack and define macro-level system boundaries (e.g., microservices, monolithic, client-server).
- **Data Strategy:** Outline high-level data flows, integration points, and external dependencies.
- **Technology Selection:** Finalize programming languages, frameworks, cloud infrastructure, and database types.

2.Detailed / Low-Level Design (LLD):Micro-level technical specifications.

Translate macro architecture into fine-grained build specifications.

- **Database Schema:** Construct Entity-Relationship Diagrams (ERDs), table structures, indexing strategies, and data relationships.
- **API & Module Design:** Define class diagrams, data contracts, input/output parameters, and REST/GraphQL endpoint schemas.
- **UI/UX Specs:** Create wireframes, user journeys, interactive prototypes, and design system design assets.

3.Security, Performance & Operations Planning:System quality attributes & safeguards.

Address non-functional requirements to ensure safety, reliability, and scale.

- **Security Framework:** Design authentication/authorization workflows, encryption policies, and data privacy protocols.
- **Infrastructure Blueprint:** Define deployment environments, CI/CD pipelines, auto-scaling parameters, and failover/recovery plans.

4.Verification & Prototyping:Technical feasibility testing.

Validate design assumptions before committing development resources.

- **Proof of Concept (PoC):** Build isolated code samples or prototypes to test risky or novel integration points.
- **Design Reviews:** Run multi-disciplinary reviews across engineering, product, QA, and security leads to catch design gaps.

5. Sign-Off & Development Handshake: Finalizing baseline deliverables.

Formally baseline the design deliverables and prepare for implementation.

- **Work Breakdown Structure (WBS):** Refine development tasks, effort estimates, and sprint breakdowns based on detailed specs.
- **Stakeholder Sign-Off:** Secure approval on the Software Architecture Document (SAD), final UI mocks, and operational budgets.